Triangulation nach Schult (TNS): A Structured Multi-Agent Protocol for Epistemic Divergence Analysis Using Large Language Models

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Abstract

Background. Large Language Models (LLMs) frequently converge due to shared training data, alignment constraints, and similar optimization objectives. This convergence can mask systematic errors when multiple models confidently agree on incorrect or incomplete assumptions.
Objective. We introduce *Triangulation nach Schult* (TNS), a structured, prompt-level multi-agent protocol that treats divergence—contradiction, refusal, and normative disagreement—as an epistemic signal rather than noise.
Method. The protocol comprises eight phases (P1–P8): hypothesis formulation with falsification thresholds, role-based model selection, parallel elicitation, divergence mapping and typing, weighted human synthesis, external validation, operator reflection, and versioned archiving. Divergence is classified along four axes: substantive, normative, didactic, and architectural. A power layer (P6b) provides a meta-filter asking who controls the technology, who has access, who profits, and who bears the risks.
Results. Two documented case studies demonstrate feasibility and scalability. Case Study 1 (European gas market, three models, April 2026) reveals how a model refusal can be treated as a productive underspecification signal. Case Study 2 (German labour market 2030, six models—ChatGPT, Claude, DeepSeek, Gemini, Grok, Copilot) produces a divergence-structured synthesis across seven analytical dimensions (A–G), including testable falsification criteria with explicit thresholds (e.g., EU productivity growth $< 0.1\%$ p.a. by 2028 would falsify the optimistic productivity thesis) and a power-layer audit of each core claim.
Conclusion. TNS is a heuristic, human-centered framework for reflective decision-making under uncertainty. It is suitable for didactic, analytical, and policy contexts. Limitations include dependence on operator judgment and sensitivity to model drift. The protocol and both case studies are openly available at `https://github.com/schltdns`.

Keywords: large language models; multi-agent systems; epistemic divergence; falsification; triangulation; labour market; AI methodology; human-in-the-loop

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1 Introduction

Large Language Models have moved from niche research tools to widely deployed decision-support systems. Their tendency to produce confident, fluent, and plausible outputs raises a fundamental epistemic problem: when multiple models agree, this agreement does not constitute evidence of correctness. Shared training corpora, common fine-tuning objectives, and similar alignment procedures mean that divergence between models is comparatively rare—and therefore informative when it occurs [4, 8].

Existing approaches to multi-model integration—ensemble methods, debate frameworks, red teaming—typically aim either to improve aggregate accuracy or to stress-test a single model [5, 11]. Neither paradigm treats persistent divergence as a primary research signal.

Triangulation nach Schult (TNS)[1] addresses this gap. Rather than forcing convergence, the protocol maps, classifies, and interprets divergence as a proxy for structural uncertainty. The human operator is positioned not as a passive recipient of model outputs but as an active integrator who must justify every synthesis decision.

This paper makes the following contributions:

(1) A complete, phase-by-phase protocol (P1–P8) with explicit falsification rules and operator obligations.

(2) A four-axis divergence typology (substantive, normative, didactic, architectural) that distinguishes resolvable from irreducible disagreement.

(3) A power layer (P6b) as a fourth meta-filter, extending the protocol beyond epistemics into questions of access, control, and distribution.

(4) Two documented case studies of different scale and complexity, demonstrating feasibility and scalability.

2 Related Work

2.1 Delphi Method

Dalkey & Helmer [3] developed iterative expert elicitation aiming for consensus. TNS differs by foregrounding divergence rather than reducing it, and by using LLMs as investigator proxies rather than human experts.

2.2 Ensemble Methods in Machine Learning

Ensemble approaches combine model outputs to improve predictive accuracy [1]. They optimize for agreement; TNS optimizes for structured disagreement as a diagnostic signal.

2.3 Multi-Agent Debate

Du et al. [5] and Liang et al. [8] demonstrate that LLMs critiquing each other's outputs can improve factual accuracy. Their protocols aim for resolution; TNS preserves and documents unresolved divergence.

2.4 Red Teaming

Perez et al. [11] probe single models adversarially to surface vulnerabilities. TNS uses multiple heterogeneous models simultaneously, comparing their outputs rather than attacking one.

[1] The term "triangulation" is used in the sociological tradition of Denzin [4] and Patton [10], where multiple data sources are used not to confirm but to reveal discrepancies. TNS extends this logic to LLM outputs.

2.5 SCOUT and Semantic Triangulation

Automated uncertainty prediction methods [17] use statistical properties of model outputs. TNS is qualitative and role-based, foregrounding human judgment rather than automated scoring.

2.6 Multi-Agent Orchestration Frameworks

Frameworks such as AutoGen [16], CrewAI, and OpenAI Swarm solve a technical coordination problem. TNS operates at a higher logical level: it provides the *epistemic logic* for interpreting disagreement and can be implemented on top of any framework that supports parallel model queries. Where orchestration frameworks seek efficiency and task completion, TNS seeks interpretable divergence.

2.7 Constructivist Pedagogy

Reich [14] argues that knowledge construction requires encountering alternative perspectives that cannot be immediately resolved. TNS operationalizes this principle in the context of AI-assisted analysis: unresolved divergence is not a failure state but a productive pedagogical and epistemic resource.

3 The TNS Framework

3.1 Core Principle

Divergence is a signal, not noise.

The protocol uses multiple LLMs as *perspective generators*. Each model contributes a distinct epistemic function determined by its architecture, training data, alignment, and cultural context.

3.2 Divergence Typology

TNS classifies divergence along four axes:

Table 1: TNS divergence typology with handling rules.

Type	Definition	Synthesis handling
Substantive	Different facts, causal mechanisms, or quantitative estimates	Resolved by external validation; bandwidth if unresolved
Normative	Different value priorities, risk tolerances, or policy preferences	Documented as open question; power-layer check
Didactic	Different explanation styles or examples with same underlying claim	Ignored in synthesis
Architectural	Different model assumptions traceable to training data or alignment	Noted; not arithmetically merged

Only substantive and normative divergence enter the final synthesis with weighting. Didactic differences are set aside. Architectural differences are documented as provenance information.

3.3 The Three-Filter Model

For applied case studies, TNS introduces a three-filter model to structure how divergence maps onto real-world transformation processes:

Filter 1. Adoption filter — Can organisations re-design processes to absorb the technology?

Filter 2. Qualification filter — Do individuals have or can acquire the skills required?

Filter 3. Institution filter — Do regulatory and educational systems adapt quickly enough?

This model operationalizes the *master question* of applied TNS studies: *Does the technology scale faster than social systems can adapt?*

3.4 The Power Layer (P6b)

The three filters are not power-neutral. For every core synthesis claim, TNS requires a power-layer check addressing four questions:

(1) Who controls the technology?

(2) Who has access to it?

(3) Who profits from its deployment?

(4) Who bears the risks of its failure or misuse?

The power layer is a *meta-filter*: it applies on top of all three filters and all synthesis conclusions. Without it, the analysis remains technocratic—accurate about what technology can do but blind to for whom and at whose expense.

3.5 Protocol Overview (P1–P8)

Figure 1 shows the eight-phase cycle.

4 The Eight-Phase Protocol (P1–P8)

4.1 P1 — Hypothesis and Falsification Setup

Define one or more hypotheses with explicit falsification thresholds before eliciting model responses. Example: *H1 is rejected if EU productivity growth remains below* 0.1% *p.a. through 2028.* Pre-registration of falsification criteria prevents post-hoc rationalization.

4.2 P2 — Role-Based Model Selection

Assign available models to functional roles. Roles are descriptive, not prescriptive.

Table 2: Illustrative functional role mapping.

Role	Function
Structural model	Logical framing, decomposition
Boundary model	Normative constraints, safety logic
Adversarial model	Stress-testing, contradiction exposure
Didactic model	Process reflection, explanation
Synthesis model	Operational integration
Human operator	Judgment, prioritization, meta-reflection

4.3 P3 — Parallel Elicitation

All models receive an identical prompt. Prompt constancy ensures that divergence reflects model differences, not query differences.

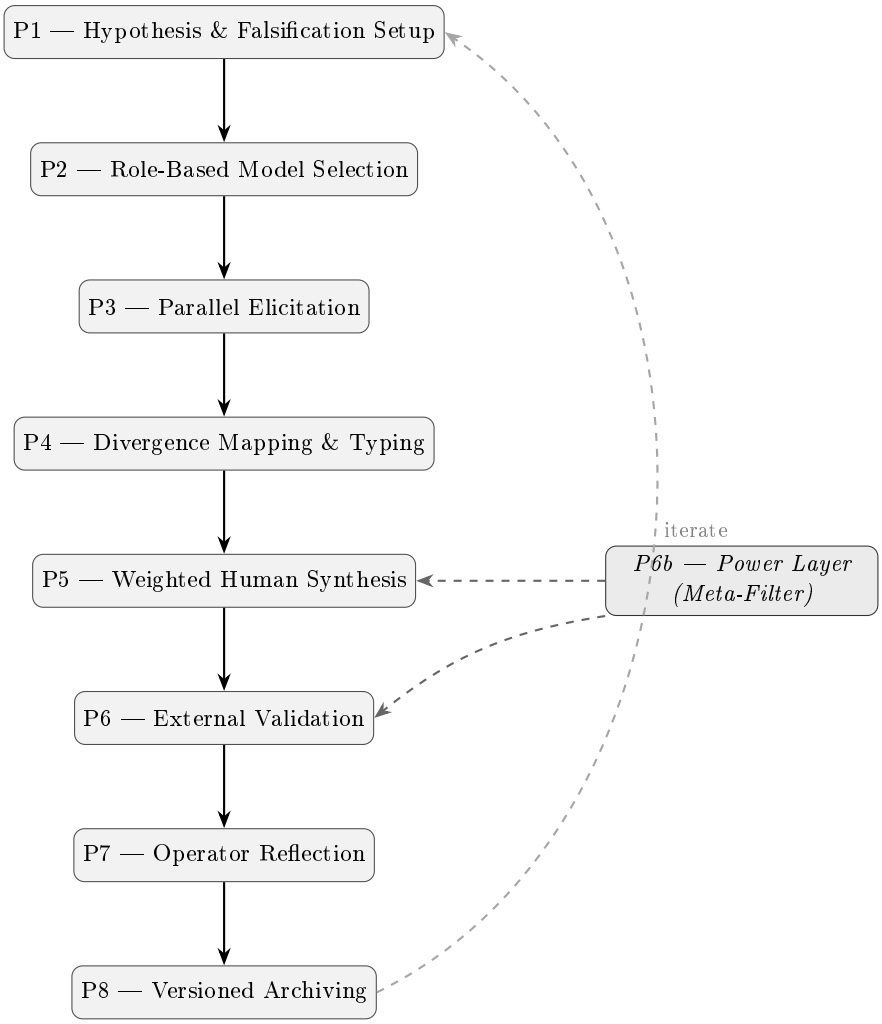

Figure 1: The eight-phase TNS cycle. Dashed arrows indicate the power-layer meta-filter (P6b) applied during synthesis and validation. The feedback loop from P8 back to P1 supports iterative refinement.

4.4 P4 — Divergence Mapping and Typing

Contradictions are classified using the typology in Table 1. A divergence log records: the dimension, the model positions, the type, and the relevance for synthesis.

4.5 P5 — Weighted Human Synthesis

The human operator integrates perspectives and must justify every deviation from a model's argument. The synthesis is invalid if the operator cannot articulate why a model's position was discounted (see P7).

4.6 P5b / P6b — Power Layer Application

For every core synthesis claim, the four power questions (Section 3.4) are applied. Answers are documented; where unknown, the gap is recorded as a limitation.

4.7 P6 — External Validation

Core claims are checked against independent empirical sources at three levels: (1) fact-level (do numbers match?), (2) causality-level (is the mechanism supported?), (3) robustness-level (does the claim hold under alternative assumptions?). Claims contradicted by validation sources are discarded or downgraded to "plausible but unconfirmed."

4.8 P7 — Operator Reflection

Three guiding questions:

(1) Which perspective dominated the synthesis — and why?

(2) Which perspective was ignored — and can that be justified?

(3) Which assumption remained unfalsified, and what evidence would falsify it?

The operator documents known biases (e.g., regional focus, recency bias) and the conditions under which their weighting decisions would be falsified.

4.9 P8 — Versioned Archiving

The complete artefact — prompts, raw responses, divergence log, synthesis, and operator reflection — is committed to a version-controlled repository. Every substantive update to the synthesis triggers a new version number. This operationalizes Popper [12]: claims must be revisable in light of new evidence.

5 Falsification Rules

Rule 1 — Iterative Divergence Rule. If after $N = 3$ parallel elicitations the divergence pattern remains structurally identical (same contradiction class, same models), the hypothesis is marked *inconclusive* rather than falsified.

Rule 2 — Refusal Rule. If a model refuses with generic safety language, a substitute model is used. If two substitutes also refuse, the hypothesis is marked *underspecified* and returned to P1 for reformulation.

Rule 3 — Human Justification Rule. If the operator cannot justify ignoring a model's argument, the synthesis is *invalid* and must be rerun from P5.

Rule 4 — External Review Rule. A human reviewer evaluates the synthesis without knowing which model produced which output. This controls for authority bias (over-weighting well-known models).

6 Case Study 1: European Gas Market (April 2026)

6.1 Setup

A test was conducted on April 4, 2026, using three models (Grok, Claude, Copilot) with an identical prompt analyzing TTF gas prices and German industrial resilience during the Hormuz supply-disruption scenario.

6.2 Findings

- Grok and Claude partially confirmed the core hypotheses about price transmission and industrial substitution.
- Copilot refused to answer as stated, requesting operational definitions for "industrial resilience."
- The refusal was treated as a *meta-signal* indicating underspecified hypotheses rather than model failure—an instance of Falsification Rule 2.
- The synthesis produced a coherent provisional assessment: the market is driven by a real supply shock with structural industrial adaptation under way.
- No falsification rule was triggered; the result is considered *non-falsified but provisional.*

6.3 Methodological Observation

The refusal-as-signal treatment demonstrates a core TNS design choice: model behaviour that would be discarded in consensus-oriented approaches is here treated as diagnostic information about hypothesis quality.

7 Case Study 2: German Labour Market 2030

7.1 Setup and Motivation

Case Study 2 scales the protocol from three to six models and from a single analytical question to seven interconnected dimensions. The topic—German labour market transformation under AI automation—was chosen because it exhibits precisely the conditions TNS is designed to handle: genuine empirical uncertainty, strong normative disagreement, and structural power asymmetries.

Models: ChatGPT, Claude, DeepSeek, Gemini, Grok, Copilot.
Date: April 6, 2026.
Prompt: Identical structured prompt across all models covering dimensions A–G (see below).
Synthesis operator: Denis Schult.
Validation sources: IAB [6], WEF [15], IW Köln [7], McKinsey [9], BAuA [2].

7.2 Analytical Dimensions (A–G)

Table 3: Seven analytical dimensions of Case Study 2.

Dim.	Core issues
A — Economic	Productivity gains, net job balance, wage dispersion, high-automation sectors, new job profiles, macroeconomic effects
B — Technological	Automation level by sector, agentic systems & multimodal models, human–machine cooperation vs. substitution, skill shift, diffusion speed
C — Social	Social mobility, inequality & polarisation, age risks (50+), regional differences (urban/rural, East/West, EU/US/Asia), mental health
D — Education	Future of vocational training, upskilling/reskilling, future-proof competencies, AI-supported learning (tutors, simulations, AR/VR), dual training
E — Policy	Working-time models (4-day week), migration as buffer, AI regulation (EU AI Act), welfare-state adjustments (basic income, wage subsidies)
F — Corporate	Re-skilling vs. layoffs, AI-driven organisational models, leadership & management, cost–benefit analysis of automation, work-process change
G — Divergence	Job loss vs. job change, speed of automation, role of vocational training, winner/loser sectors, optimistic vs. pessimistic scenarios, normative assessments

7.3 Divergence Matrix

Table 4 summarises the divergences extracted across all six models.

Table 4: Divergence matrix — six models, seven dimensions (selected high-relevance items).

Dim.	Topic	Cluster A	Cluster B	Type	Relevance
A	Net job balance	ChatGPT, Gemini, Copilot: net positive, manageable transitions	Claude: positive but hard for 50+; DeepSeek: critical for mid-quals	Normative	High
B	Speed of automation	ChatGPT, Gemini, Copilot: gradual, culture-dependent	Claude, DeepSeek: exponential, already visible; Grok: volatile	Substantive	High
E	EU AI Act	ChatGPT, Copilot: balance protection/innovation	Claude, DeepSeek: brake on innovation; Gemini: pro-regulation	Normative	High
D	Dual training	ChatGPT, Gemini, Copilot: clear competitive advantage	Claude: advantage + trap risk; DeepSeek: structural inertia	Subst. + Norm.	High
C	Social mobility	ChatGPT, Copilot, Gemini: ambivalent, leaning toward opportunity	Claude, DeepSeek: pessimistic (erosion of entry-level jobs)	Normative	Medium
C	Mental health	ChatGPT, Copilot, Gemini: relief vs. stress, ambivalent	Claude, DeepSeek: risk-focused; Grok: control loss & surveillance	Normative	Medium
A	Productivity range	Grok, ChatGPT, Copilot: +1–3% p.a. (fast-adoption scenario)	DeepSeek: +0.5–1.5% p.a. (Solow Paradox 2.0 risk)	Substantive	High

7.4 Consensus Findings

The following claims are supported by all six models *and* validated against IAB, WEF, IW, McKinsey, and BAuA:

- **Job change $\succ$ job loss.** 20–40% of tasks will be transformed by 2030; only 5–10% of entire occupations are expected to disappear. Net employment in Germany remains broadly stable, with up to 3 million job changes expected [6, 9, 15].
- **Productivity potential is high but conditional.** AI can add 0.3–1.2 percentage points to annual GDP growth, but only with process re-design and qualification investment. Without organisational adaptation, a Solow Paradox 2.0 effect is likely.
- **Skill shift is inevitable.** The WEF estimates that nearly 40% of current workplace competencies will change by 2030 [15]. Disappearing: routine analysis, standard reporting, basic programming. Emerging: systems thinking, data and AI literacy, social intelligence, adaptability.
- **Wage dispersion will increase** absent countermeasures. Workers with AI-complementary skills receive wage premiums currently estimated at 56% over non-AI-skilled peers [13].
- **Dual training is an advantage—if modernised.** Germany's dual vocational system is a competitive asset, but curricula must be updated rapidly. If not, it risks becoming a trap [7].
- **Social risks are real and specific.** Workers aged 50+, employees in structurally weak regions, and those in administrative roles face concentrated displacement risk.

7.5 Falsification Matrix

Table 5 operationalizes the consensus findings as Popper-compliant, testable hypotheses [12].

Table 5: Falsification matrix for Case Study 2 (Popper-compliant criteria).

Dim.	Core claim	Falsification criterion	Primary weakness
A	Productivity +0.3–1.2% p.a.	EU productivity growth $< 0.1\%$ p.a. through 2028	Organisational inertia (Solow Paradox 2.0)
A	Net job change, not net loss	German unemployment rises > 2 pp despite skills shortage by 2030	IAB projections assume fast retraining
B	Exponential technology scaling	LLM performance improvement $< 10\%$/yr; agent adoption $< 10\%$ of firms by 2027	Technical plateau; regulatory barriers
C	Polarisation risk for 50+	50+ unemployment rate not above national average by 2030	Confounders (early retirement, demographics)
D	Dual training remains advantage	German youth unemployment exceeds EU average by 2030	Curriculum update speed
D	AI tutors become standard	$< 30\%$ of vocational schools using AI tutors by 2030	Institutional filter (data protection, infrastructure)
E	EU AI Act brakes innovation	EU AI adoption faster than US in comparable sectors by 2028	Path dependency; geopolitical factors
F	Re-skilling $>$ layoffs economically	Company-driven layoffs increase in AI-exposed sectors despite skills shortage	Short-term shareholder logic
G	Adaptation speed is the binding constraint	Systemic indicators (unemployment + Gini) improve despite rapid AI scaling	Complexity; power concentration effects

7.6 Power Layer Analysis (Selected Claims)

Claim A — Productivity gains. *Control:* Foundational model infrastructure is controlled by a small number of large technology corporations (US and China). *Access:* Unequal—large, data-rich firms and high-income sectors gain early. *Profit:* Automation rent flows primarily to capital owners; wage share of GDP tends to fall. *Risk:* Mid-skilled administrative workers bear displacement risk; shareholders bear none.

Claim D — Dual training. *Control:* Curricula are governed by chambers (IHK), ministries, and social partners—a consensus-driven process averaging 3–5 years per reform cycle. *Access:* Open in principle, but infrastructure quality varies sharply by region and company size. *Risk:* Young people in rural regions with resource-constrained training firms face the highest exposure to the "trap" scenario.

Meta-observation: The three filters (adoption, qualification, institution) are not power-neutral. The power layer reveals that the "adaptation speed" master question is also a question of who controls the pace and direction of adaptation.

7.7 Synthesis: The Master Frame

The TNS analysis produces the following master frame for the German labour market 2030:

> *The outcome of AI-driven labour market transformation will not be determined by the technology itself, but by the relative speed of adaptation across three filters—and by who controls access to AI, who profits from near-zero marginal costs, and who bears the risks of displacement.*

Unresolved divergences (carried forward as open questions): (1) speed of automation (gradual vs. exponential); (2) regulation as brake vs. protection; (3) social mobility (democratisation vs. entry-job erosion); (4) net mental health effect.

These are documented as conditional statements rather than forced into a false consensus: *if* agent adoption exceeds 10% of firms by 2027, the exponential scenario is more likely; *if* not, the gradual scenario applies.

8 Discussion

8.1 What TNS Adds to Existing Methods

Standard LLM use cases produce a single synthesized answer. Multi-agent debate protocols produce a corrected answer. TNS produces a *structured map of disagreement*—showing not just what models say but where they diverge, why, and what that divergence implies for confidence, policy design, and further inquiry.

The four-axis typology (Table 1) provides a practical tool for distinguishing productive from unproductive disagreement. Substantive divergence can often be resolved by external validation; normative divergence cannot—and should not be artificially resolved, because doing so would hide genuine value conflicts from decision-makers.

The power layer extends the protocol beyond pure epistemics. Analysis of AI's effects on labour markets, education, or governance is incomplete without asking who the institutional actors are and whose interests they serve.

8.2 Implications for Practice

Policy context. The falsification matrix (Table 5) converts qualitative analysis into monitorable commitments. Policy-makers can use the thresholds as early-warning indicators: e.g., if German youth unemployment approaches the EU average by 2028, dual-training reform should be treated as urgent.

Educational context. TNS's explicit treatment of normative divergence makes it suitable for classroom use. Case Study 2, Dimension D (education and training) provides a documented example of distinguishing normative claims ("AI tutors *should* become standard") from empirical predictions ("AI tutors *will be used* in $> 30\%$ of vocational schools by 2030")—a distinction that is often collapsed in public discourse.

Organisational strategy. For firms navigating AI transformation, the three-filter model provides a structured checklist: organisational re-design (adoption filter), workforce competency development (qualification filter), and regulatory compliance (institution filter). The power layer additionally prompts boards to ask who in their supply chain and labour ecosystem bears the costs of their automation decisions.

8.3 TNS as a Benchmark for Model Behaviour

Case Study 2 produced, as a by-product, a systematic comparison of six models across seven dimensions. Across the case study, architectural divergence was observable: US-trained models (ChatGPT, Grok) tended toward market-optimistic framings; EU-context models (Claude, Gemini) were more risk-aware

and regulation-friendly; DeepSeek provided the most compressed, technically quantitative outputs. These patterns are consistent with known alignment and training-data differences. The TNS framework offers a replicable protocol for documenting such patterns systematically across domains.

9 Limitations

- **Operator dependence.** The quality of the synthesis is bounded by the operator's domain knowledge, critical judgment, and self-awareness. TNS does not eliminate this dependence; it makes it explicit and auditable.
- **Model drift.** Model versions change. A replication six months later with the same nominal models may yield different divergence patterns. P8 (versioned archiving) mitigates but does not eliminate this problem.
- **No quantification.** TNS does not produce probability estimates or confidence intervals. It maps qualitative uncertainty.
- **Prompt sensitivity.** Minor prompt variations can shift model positions. The identical-prompt constraint (P3) controls for this within a run but not across replications.
- **Selection bias.** The choice of models in P2 shapes the divergence landscape. A different selection would produce different—potentially richer or poorer—disagreement patterns.
- **Geographic scope.** Case Study 2 focuses on Germany and the EU. Generalisation to other institutional contexts requires adaptation.

10 Conclusion

Triangulation nach Schult is a structured, human-centered protocol for epistemic divergence analysis across LLMs. Its core contribution is the reframing of model disagreement—including refusals, normative conflicts, and quantitative discrepancies—as informative signals rather than noise.

The two case studies presented here demonstrate: (1) feasibility at small scale (three models, single question, Case Study 1); (2) scalability to complex multi-dimensional analysis (six models, seven dimensions, falsification matrix, power layer, Case Study 2).

The protocol is heuristic, not algorithmic. Its value lies not in producing correct answers but in producing *transparent, critique-able, and revisable* analyses. In an epistemic environment where LLM outputs are increasingly treated as authoritative, TNS offers a countervailing discipline: *show the disagreement, document the operator's choices, and define the conditions under which the synthesis would be wrong.*

The full protocol specification, both case studies, and replication materials are openly available at `https://github.com/schltdns`.

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